

Mukul Aggarwal

Software Engineer

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EXPERIENCE

Designing Freelancer

2022 & 2024

Self Employed

- Designed professional, visually engaging digital assets for clients across social media, business, education, and marketing domains. Collaborated directly with clients to understand their requirements and transform ideas into modern, brand-consistent designs while ensuring timely delivery.

Tech: Canva, Figma

Junior Dev | Designer

2026

Rootpay

- At Rootpay, a US-based fintech startup, I contributed as a Frontend Engineer, building secure, performant, and user-friendly web experiences. My work focused on strengthening client-side security through anti-tampering techniques and frontend security testing, while continuously refining the user interface for a seamless experience. I also optimized the product's SEO and Generative Engine Optimization (GEO), improving its discoverability across both traditional search engines and AI-powered search platforms. Working in a fast-paced startup environment taught me how to deliver scalable solutions where security, performance, and user experience are equally important.

Tech: Next JS, Canva, React, Cyber Security, Stable Coin, Crypto

Junior Dev

2026

DataDensity

- At Datadensity, an Indian Physical AI startup, I contributed to the development of data collection and visualization solutions for AI training pipelines. The company focuses on generating high-quality multimodal datasets by combining IMU sensor data and camera footage to train mechanical robots for 21-point keypoint annotation and motion understanding. I worked on building user-facing interfaces, improving data workflows, and supporting tools that enabled efficient collection, processing, and visualization of training data for robotics and computer vision applications. This experience deepened my understanding of Physical AI, sensor fusion, and the challenges involved in creating reliable datasets for machine learning systems.

Tech: Next JS, Cyber Security, IOT, Embedded System, Python

Technical Coordinator

2026

AIML hub of IIIT

- As the Technical Coordinator at AIML Hub, IIIT, I lead technical initiatives for a community of 100+ AI and ML enthusiasts. I organize workshops, coding sessions, and technical events while collaborating with fellow coordinators to build an engaging learning environment. Through mentorship, knowledge-sharing sessions, and project-driven activities, I help students strengthen their technical skills and stay updated with advancements in Artificial Intelligence and Machine Learning. This role has enhanced my leadership, event management, and technical communication skills while contributing to a vibrant student developer community.

Tech: Machine Learning

EDUCATION

I am sorry can't reveal the name

2008 – 2023

Schooling till 12th

My worst experiences of life.

Jaypee Institute of Information Technology

2024 – 2028

B-Tech Computer Science

Ongoing.

PROJECTS

Naughty Pikachu

github.com/me-is-mukul/naughty-Pikachu

A lightweight real-time chat application built to experiment with WebSockets, enabling instant messaging without authentication while supporting temporary local message storage.

Tech: JavaScript, WebSocket, Node.js, HTML, CSS

BudgetPe

github.com/me-is-mukul/BudgetPe

An AI-powered personal finance platform that analyzes transaction history, categorizes expenses, and provides personalized budgeting insights to help users improve their financial habits.

Tech: JavaScript, React, Node.js, AI, MongoDB

Laser Communication IoT

github.com/me-is-mukul/Laser-Communication-IOT-

An IoT-based communication system demonstrating optical data transmission using laser modules and optical fibers, showcasing the practical implementation of laser-based communication principles.

Tech: C++, Arduino, IoT, Laser Communication, Embedded Systems

KIWI

github.com/me-is-mukul/KIWI

A Git-inspired version control system built entirely in Java. Features content-addressable object storage, staging, commits, branching, history tracking, and repository management while demonstrating the core internals of modern VCS tools.

Tech: Java, File System, SHA-1, CLI, Version Control

Time Complexity Beauty

github.com/me-is-mukul/Time-Complexity-Beauty

A research-oriented project that validates the practical performance of Master Theorem by benchmarking the Rabin-Karp algorithm across datasets of varying sizes and visualizing empirical time complexity.

Tech: LaTeX, Python, Algorithms, Rabin-Karp, Research

Jolly Joysticks

github.com/me-is-mukul/jolly-joysticks

A cross-platform desktop client developed in Python for integrating multiple AI models into a unified interface, providing an efficient and seamless workflow across Windows, Linux, and macOS.

Tech: Python, PyQt, AI APIs, Desktop Application, Cross Platform

SKILLS

Backend	Node.js, Express.js, FastAPI, REST APIs
Cloud & Deployment	Render, Railway, AWS, Vercel
Cybersecurity	OWASP, Burp Suite, Wireshark, JWT, OAuth, Client-side Security Testing
DevOps & Tools	Docker, Docker Compose, GitHub Actions, Nginx, Linux, Git
Database	MySQL, MongoDB, Firebase
Programming Languages	C++, Python, Java, JavaScript, TypeScript, SQL
Machine Learning	Python, PyTorch, OpenCV, NumPy, Pandas
Frontend	React, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS
Tools & Design	GitHub, Figma, Canva, Postman, VS Code, Three.js